



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

From Geometric Documentation to Knowledge Integration: A Web-Based Approach to Lightweight Point Cloud Management and Interaction for Historic Buildings

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Abstract

With the widespread application of 3D scanning in heritage conservation, point cloud data have become a digital carrier for historic buildings. However, datasets often reach GB to TB scales, causing slow loading and high hardware dependency, which limits the effective integration of point cloud data into practical conservation workflows (Fawzy, 2019). Current practice relies heavily on multiple specialised software packages and high-performance local computing environments. This fragmented workflow creates significant information barriers across data acquisition, processing, and application, hindering the efficient circulation of point cloud data throughout the conservation process (Roe, 2025). Furthermore, enabling the sharing of point cloud data while ensuring data security remains a practical challenge.

In response to these issues, this study proposes a web-based approach that combines lightweight point cloud processing with interaction design, aiming to develop an integrated online platform for storage, visualisation, analysis, and management (Figure 1). The approach also explores a paradigm shift in digital heritage conservation from "geometric documentation" to "knowledge integration".

At the platform architecture and interaction level, this platform adopts a dual-database model: the complete point cloud dataset is retained on the local server, while a lightweight database is deployed on the cloud, with both databases sharing the same coordinate system to ensure seamless coordination. The platform is built on a SaaS architecture, allowing users to access the cloud database directly through a standard web browser

without installing specialised software (Figure 2). During this process, users can perform interactive operations such as browsing, zooming, rotating, sectioning, annotating, and measuring, but do not access the complete raw data, thereby ensuring data security without compromising usability. Additionally, the platform supports differentiated functional organisation and access tiering for various usage scenarios.

The realisation of the above architecture and interaction functions relies on the following technical pathways. The platform is built upon WebGL and 5G communication infrastructure, employing voxel downsampling (Shi & Luo, 2024) and octree-based multi-resolution partitioning strategies (S. Huang, 2023) for data preprocessing. The rendering engine, developed with Three.js, implements dynamic on-demand loading, view-dependent Level of Detail (LOD) control, and GPU-accelerated rendering (Putra et al., 2023), significantly enhancing rendering performance and interaction fluidity while maintaining high visual precision (Uslu & Uysal, 2023).

Taking the "Guangzhou Historic Building Information System" developed based on this platform as a case study, the platform has deployed a differentiated permission management mechanism in practice. The public-facing interface provides lightweight point cloud browsing and interaction functions, lowering the access threshold for non-expert users; the management-facing portal grants full database management and slice export permissions to support conservation and restoration planning and decision-making. Practice demonstrates that, built on lightweight sharing, the platform promotes the transformation of raw survey data into knowledge resources that can serve conservation decision-making through structured organisation, access tiering, and interaction design.

Compared with existing web-based 3D viewing tools such as DFG-3D-Viewer, this platform integrates multiple types of data within a single permissioned web environment, aiming to build comprehensive infrastructure covering digital surveying, heritage management, professional collaboration, and public dissemination. Current limitations include scalability for extremely large datasets; future work will focus on deeper integration with the IFC data standard (Aydın, 2025) to enhance interoperability in architectural heritage data.



View point cloud models

Export point cloud slices

One-click sharing

Project management

Fig. 1 Overview of the Web-Based Platform for Point Cloud Visualisation, Interaction, and Management.

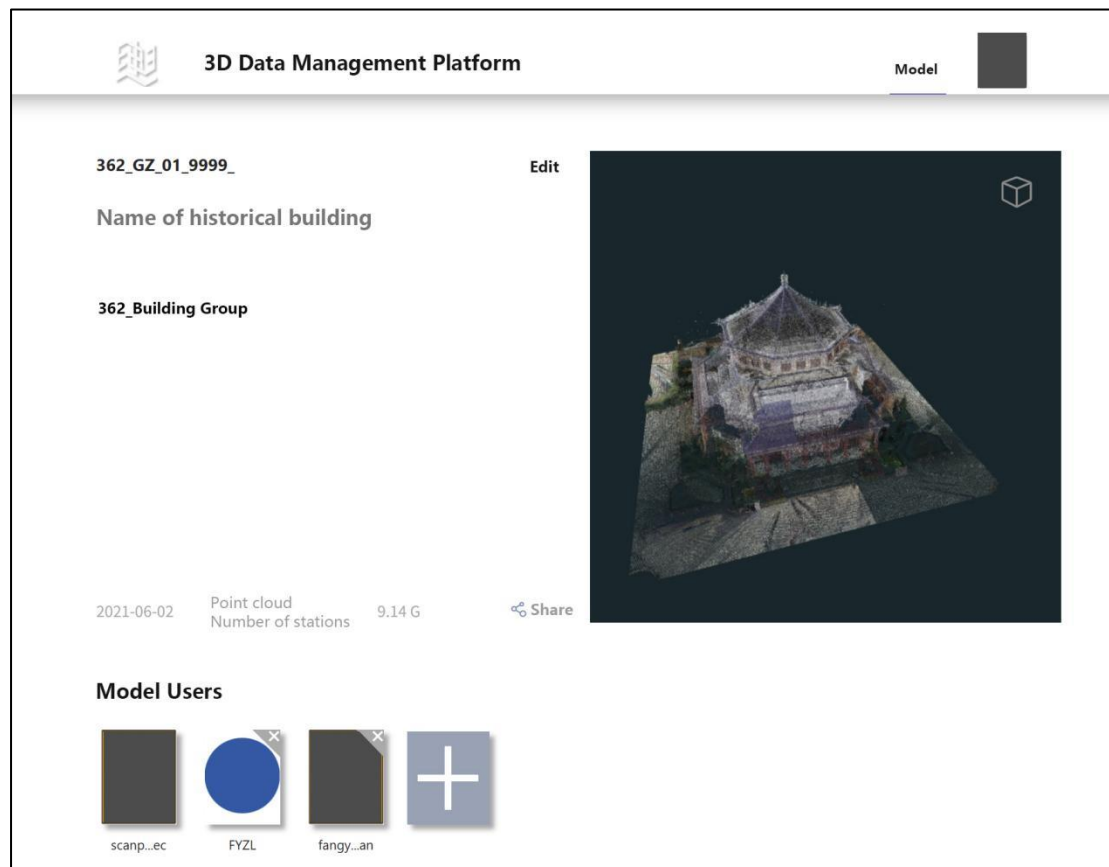


Fig. 2 3D Data Management Platform Model Management and Basic Information Interface.

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